# REAL-TIME ACCIDENT MONITORING SYSTEM USING IOT

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**Abstract**

This project involves the development of an IoT-based real-time accident monitoring system for enhanced accident analysis. The system integrates multiple sensors to continuously capture and log critical vehicular parameters, including speed, acceleration, geolocation, and environmental conditions. Upon detecting an accident, it autonomously activates an alert mechanism through a GSM module, transmitting notifications to emergency responders and predefined contacts. Simultaneously, it uploads detailed incident data to the cloud platform ThingSpeak, where it can be accessed for comprehensive analysis. The system's ability to store and transmit real-time data enables accurate accident reconstruction, identifying factors such as driver behavior, vehicle dynamics, and external conditions contributing to the incident. If the driver consumes alcohol, then the system stops the engine so that accidents due to alcohol consumption will be reduced. The collected data also facilitates post-accident investigations and helps improve safety measures. Additionally, the system supports remote diagnostics and real-time vehicle monitoring, providing valuable insights for enhancing road safety. This integrated approach aims to reduce emergency response times, improve the accuracy of accident investigations, and promote safer driving practices.

**Keywords:** Emergency Alerts, Drunken Driving Detection, GSM, Speed Monitoring, Acceleration, Over-Braking, Tilt Detection.

# 1. Introduction

## 1.1 Objective

The Internet of Things (IoT) has evolved into a transformative technology that revolutionizes the way we connect with and interact with the world around us. Through the connection of computer devices embedded in existing Internet infrastructure, IoT provides unprecedented connectivity, enabling seamless communication between devices, systems, and services. This connectivity goes far beyond traditional communication with machine-to-machine (M2M) interactions and includes a variety of protocols, domains, and applications. In virtually every automation area, IoT has paved the way for innovative solutions and advanced applications, unlocking new opportunities for efficiency, productivity, and convenience. From optimizing energy distribution to intelligent networks and transforming healthcare with cardiac monitoring implants, IoT has become an integral part of modern life. One particularly promising application of IoT technology lies in the field of vehicle security. The spread of intelligent objects and embedded sensors has enabled the development of sophisticated systems for accident detection and response. Smart Accident Detection and Alarm System is an advanced IoT-based solution designed to identify essential notifications for vehicle accidents and emergency services. The main goal is to improve survival rates by reducing emergency response times and ensuring prompt assistance. The system consists of multiple vehicle sensors that continuously monitor parameters such as vehicle speed, GPS coordinates, and accelerometer data. These sensor values are transmitted to a central server via wireless communication protocols, including 3G, 4G, Wi-Fi, or other available networks. The server employs machine learning algorithms to analyze incoming data and determine the occurrence of an accident. When an accident is detected, the system automatically sends an alert to emergency responders and predefined contacts, providing critical details such as location and accident severity. Additionally, notifications can be sent to victims' family members and friends via multiple communication channels, including SMS, emails, and push notifications. One major advantage of this system is its ability to function remotely, even in areas with limited or no cellular coverage. In such scenarios, satellite communications are used to ensure emergency alerts are transmitted successfully, enhancing the system's reliability in various environments. The system can also integrate with existing emergency services, enabling more efficient and coordinated responses. For instance, it can automatically notify nearby hospitals to prepare for incoming patients.

## 2. Literature survey

1. The GPS-GSM sheet, an inland navigation tracking system for automatic emergency recognition and location notification, by Shabrinet Althis, introduces a progressive vehicle tracking system (VTS) designed for domestic ships. Unlike traditional VTS used in cars, this system enables remote monitoring of ship movement and direction. It also includes real-time accident recognition features, utilizing continuous motion tracking to identify abnormal events. Once an irregularity is detected, the system immediately sends the ship's GPS coordinates to the relevant authority or owner, improving the rescue process and minimizing potential losses. The system relies on GSM communications for data transmission between onboard devices and the ground control center, limiting its use to inland waterways. However, replacing GSM with Iridium satellite communications would expand its usability to seaboard vessels, offering broader operational coverage. Additionally, this paper explores the concept of an integrated domestic waterway control system (IIWCS), incorporating this technology to enhance maritime security and navigation management.

2. Design and implementation of helmets to pursue accident zones, by Karthik P et al., focuses on improving road safety through an accident recognition system that automatically detects vehicle collisions and notifies pre-configured contacts such as family members, emergency services, police stations, and nearby hospitals. Increased urban complexity leads to worsening road conditions due to factors like traffic congestion, rising accident rates, and a high percentage of abandoned vehicles. These challenges contribute to higher transportation costs and inefficient vehicle movement. In densely populated countries such as India, delayed medical responses often result in fatalities due to a lack of timely assistance. This project aims to address these issues using GSM technology to send real-time vehicle location updates via SMS to designated recipients, ensuring immediate support. Additionally, the system offers vehicle tracking capabilities, making it particularly useful for fleet management. By regularly transmitting location data through GSM modems, businesses can efficiently monitor their vehicles, enhancing operational efficiency and security.

3. Jong-Ho Choi et al. propose a tracking methodology that integrates both traffic patterns and vehicle-specific characteristics, such as contextual background, localized positions, and vehicle measurements. A four-way overview tracking algorithm is used to determine the vehicle’s initial position and size. To improve performance under varying lighting conditions, frame difference technology is employed to reduce the impact of brightness changes on contour detection. Each vehicle is represented using four key feature points, identified through the

minimum cover distance at the vehicle's location. By aligning the detected vehicle areas, the system effectively mitigates occlusion issues and enhances tracking performance.

## 3. Existing System

People in general or crisis management have the least foggy thoughts about the extent of closure and administration. Such data is lacking, which can cause some casualties. As a result, exploration tests arise in a way that answers how a crisis vehicle can be operated to reach a misfortune spot at the right time. Therefore, it is important to guide the vehicle and regulate movement signs according to the vehicle development. Many deaths have been occurring as the outbreak into the emergency vehicle mixture has changed. To achieve these obligations, we recommend running a system called the Entra Vectice Surveillance System (EMMS). Emergency vehicle checking systems can include GPS, GSM, and GIS management. GPS devices provide information about the location of the emergency vehicle and the shortest course. The main server that allows Google Manual to provide the shortest course between crisis vehicles and mixes. Therefore, we propose another outline to control activity signals and emergency vehicle reviews and obtain the above delegation so that the crisis vehicle has the ability to surpass all traffic control signals without maintaining itself. Each movement transition has a controller that controls the activity flow. Motion control is pointed out as a button, with each button having a GSM modem (a portable-enabled global system) connected to the controller. The hub is controlled by the main server by sending tax reports to a GSM modem (a global system for portable support). As the population grows, the number of vehicles on the road increases proportionally, increasing traffic congestion, leading to an increase in corresponding road accidents. The main challenge associated with such incidents is the delayed ambulance response when it reaches the scene of the accident or when the victim is transported to a medical facility. Timely medical interventions are extremely important for improving survival. To tackle this issue, it is important to implement automated accident recognition and warning systems that immediately notify rescue personnel and facilitate rapid transport of victims to hospitals. Additionally, timely accident reporting to investigators can accelerate the testing process and reduce delays in accident analysis and response measurements.

## 4. Proposed System

This work proposes an accident detection rescue system using IoT. IoT is very useful in this aspect as it replaces the conventional monitoring systems with a more efficient scheme. The proposed system of this project is shown in Fig 4.1.

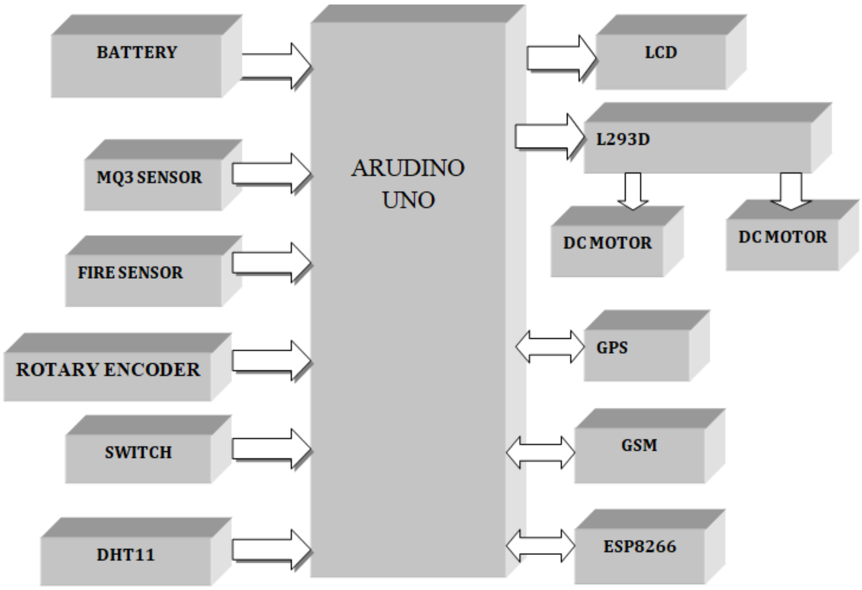

Fig 1 Block Diagram

Vehicle units include a microcontroller, MPU6050 movement sensor, MQ3 alcohol sensor, fire detection sensor, GPS module, and GSM communication module. The device is integrated into the vehicle to record accidents and transmit location data to designated recipients. The MPU6050 sensor continuously monitors sudden changes in vehicle movement, with the collected data processed by the microcontroller. The GPS module determines the real-time coordinates of the vehicle, identifies the accident location, and forwards this information to the GSM module. The GSM module then sends a warning message to pre-configured emergency contacts stored in the system, ensuring immediate notification in the event of an accident.

Flame sensors detect the presence of fire in the vehicle, while the gyro sensor monitors the alignment and angular speed of the vehicle. If any abnormal condition is detected, a warning message is displayed on the LCD (16*2), and the vehicle's location is sent to emergency contacts via the GPS and GSM modules. In critical situations, the braking system is activated immediately to ensure controlled deceleration, enhancing vehicle safety.

The Collision Anti-Crash Braking System (ACBS) is an advanced safety mechanism designed to detect potential collisions and autonomously activate the braking system. This feature slows down the vehicle or brings it to a complete stop, reducing the impact and risk of accidents. A switch is used to activate the system, ensuring the vehicle remains responsive without unnecessary sensitivity. The faster the wheel rotates, the higher the pulse frequency, indicating increased speed.

Additionally, the DHT11 sensor is used to measure the temperature inside the vehicle by reading the ambient temperature.

## 5. Hardware Components

### 5.1. ARDUINO UNO

The Arduino UNO is a widely used open-source microcontroller board, primarily developed with the Microchip Atmega328p processor by Arduino.cc. It features multiple digital and analog input/output (I/O) pins, allowing seamless integration with expansion modules and external circuits. The board includes 14 digital I/O pins and six analog input pins, providing a

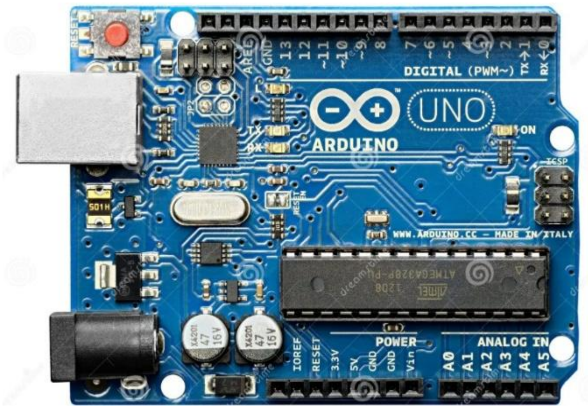


Fig: Arduino UNO

versatile interface. Programming is done using the Arduino Integrated Development Environment (IDE) via a Type B USB connection

### 5.2 LCD (LIQUID CRISTAL DISPLAY)

A liquid crystal display (LCD) is a thin, flat display device composed of either colored or monochrome pixels placed in front of a reflective or light source. Each pixel consists of two polarizing filters with perpendicular polar axes and a layer of liquid crystal molecules between two transparent electrodes. When light passes through one filter, it is blocked by the other unless twisted by the liquid crystal, allowing controlled light transmission.

Fig: LCD (Liquid Crystal Display)

### 5.3 FIRE SENSOR

A flame sensor is a type of detector designed to identify and respond to fire or flames. It plays a critical role in fire detection systems, propane and natural gas connections, and alarm systems. Industrial boilers also use flame sensors to ensure proper operation. Compared to heat or smoke detectors, these sensors respond more quickly due to their flame-detection mechanism.

Fig Fire Sensor

### 5.4 ALCOHOL SENSOR:

The MQ-3 alcohol sensor is a specialized module designed to detect the presence of alcohol in the air. It utilizes a semiconductor material containing tin oxide, which changes its electrical conductivity when exposed to alcohol molecules. As alcohol comes into contact with the sensor, it alters the sensor's resistance, which is measured by its electronic circuit. MQ-3 sensors are commonly used in breathalyzers, automobile safety systems, and alcohol detection applications. They offer advantages such as high sensitivity, fast response time, and low power consumption.

## 5.5 GSM (GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS)

The Global System for Mobile Communications (GSM) is a mobile network technology that enables mobile phones to connect to nearby cell towers. GSM networks operate on four different frequency bands. Short Messaging Service (SMS), commonly known as text messaging, was introduced as an inexpensive alternative to voice calls under GSM technology. A notable feature of GSM is the inclusion of a universal emergency number, 112, making it easier for foreign visitors to contact emergency services without knowing the local emergency number.

Fig: GSM Module

## 5.6 L2983

The L293 is a motor driver designed to provide bidirectional current flow of up to 1A with voltages ranging from 4.5V to 36V. The L293D variant supports bidirectional currents up to 600mA within the same voltage range. These motor drivers are commonly used to control solenoids, DC motors, and bipolar stepper motors, making them suitable for high-current and high-voltage applications.

## 5.7 DC MOTOR

The DC engine is designed to run on DC power. Two examples of pure DC design are Michael Faraday's homopole engine (unusual) and the engine containing the ball. This is (so far) novel. The most common DC engine types are brush and brushless types that use internal and external rectification to create alternating currents that vibrate from the DC source.

Fig: DC Motor

## 6. Results

The system is designed so that drunk driving can automatically be switched from the vehicle when the driver is deemed to be intoxicated. Additionally, in the event of a collision, the system will send the car's GPS location to the emergency contact. This means that rescuers can localize and assist passengers. When the vehicle fires, the system sends the vehicle's coordinates. This allows you to quickly determine the location of the fire and take appropriate measures. Overall, the system aims to improve traffic safety by preventing drunk driving and assisting in accidents and emergencies.

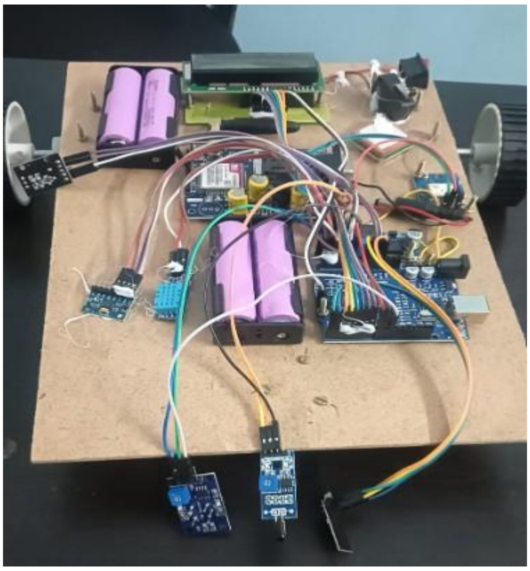

Fig: Hardware Setup

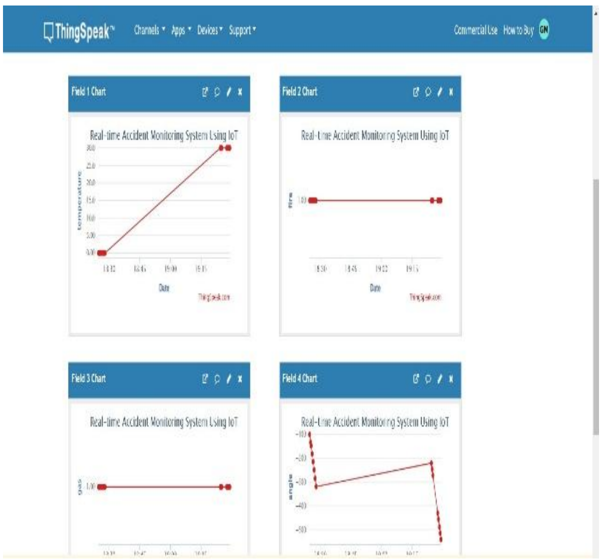


Fig: IoT platform output

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